

Supervised Learning

Supervised Learning is a type of Machine Learning where a model learns from labeled examples.

Each training example contains:

- Input data
- The correct expected answer or label

The model learns the relationship between the input and the expected output.

Two common supervised learning tasks are:

- Classification Predicting a category.

Example: Determining whether an email is spam or not spam.

- Regression Predicting a numerical value.

Example: Predicting the price of a house based on its size, location, and number of rooms.

A simple example of supervised learning is training a model to recognize cats and dogs.

The training dataset contains images labeled “cat” or “dog”. After learning from these examples, the model attempts to classify new images that it has never seen before.

The quality and accuracy of the training labels can significantly affect model performance.